

Green Operators and Zeta Functions of Recursive Factorization Systems

From Pratt Geometry to Euler Products

Orges Leka

with mathematical assistance from ChatGPT 5.6

August 2026

Abstract

A ranked recursive factorization system (RRFS) is a factorial commutative monoid whose atoms carry rank-decreasing predecessors. Its transition matrix B records the direct factors of each predecessor, and the previously used resolvent $(I - B)^{-1}$ transports ordinary root multiplicities into full recursive coordinates. We reinterpret this operator as the *Green operator*

$$G_{\mathfrak{R}} = (I - B)^{-1}.$$

The Green operator also has a dual analytic role. A local recursive energy vector ε determines root energies

$$c = G_{\mathfrak{R}}^{\top} \varepsilon,$$

and therefore a multiplicative norm $N_{\varepsilon}(h) = \exp(\varepsilon^{\top} G_{\mathfrak{R}} v(h))$. Whenever the corresponding partition sum converges, we define the zeta function of the RRFS by

$$\zeta_{\mathfrak{R}, \varepsilon}(s) = \sum_{h \in H} \exp(-s \varepsilon^{\top} G_{\mathfrak{R}} v(h)).$$

Unique factorization gives the equivalent Euler product

$$\zeta_{\mathfrak{R}, \varepsilon}(s) = \prod_{\alpha \in \mathcal{A}} (1 - e^{-s c_{\alpha}})^{-1}.$$

Thus the same analytic function has two coordinate descriptions: an Euler product in boundary coordinates and a Green representation in recursive Pratt coordinates.

For the numerical Pratt RRFS, the local energies are $\log(p/(p-1))$ and the resulting zeta function is exactly the Riemann zeta function. We then lift the numerical Pratt predecessor to nonzero prime ideals in the ring of integers $\mathcal{O}_K$ of a number field by setting $D_K(\mathfrak{p}) = (p-1)\mathcal{O}_K$ for $\mathfrak{p} \mid p$. Unique factorization of ideals makes this a genuine nonterminal RRFS, and its zeta function is exactly the Dedekind zeta function $\zeta_K(s)$. The corresponding Green operator refines the numerical Pratt tree by prime-ideal splitting and ramification. For monic polynomials over $\mathbb{F}_q$, the degree gauge is supported only on the terminal coordinate x and recovers the zeta function of the affine line, $(1 - q^{1-s})^{-1}$. For monic rational polynomials, Mahler measure gives a signed local recursive energy and leads naturally to fixed-degree and bivariate height zeta functions. We also explain terminal RRFS constructions attached to Jordan and Grassmann orbit atoms: their Green operator is the identity, but their atom multiplicities produce zeta functions with a dominant shifted Riemann-zeta factor. Finally, logarithmic differentiation yields an abstract von Mangoldt weight and links the RRFS zeta formalism to prime number laws and to Orbit Systems.

Status and scope. The classical identities for the Riemann and Dedekind zeta functions, finite-field polynomial zeta functions, Euler products, unique factorization of ideals in Dedekind domains, Mahler measure, Jordan totients, and Gaussian binomial coefficients are not new. The purpose of this paper is the RRFS synthesis: the Green operator, its dual transport of local recursive energies to root energies, and the resulting common zeta formalism. We deliberately use the word *zeta* only for the analytic partition function introduced below; no incidence-algebra zeta function is needed in this paper. The terminal Jordan and Grassmann constructions are analytic terminalizations of orbit families, not claims that a nontrivial canonical predecessor map has been found for those geometries.

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1 Introduction

Recursive factorization systems were introduced to isolate the common structure behind numerical Pratt trees, polynomial Pratt forests, and other deterministic dependency hierarchies. An RRFS has two levels of multiplicative information. The first is ordinary factorization: an object h has root multiplicities $v_\alpha(h)$. The second is recursive factorization: each atom α has a predecessor $D\alpha$, and repeated factorization of predecessors produces a finite rooted tree. The total number of occurrences of each atom in the unfolded forest gives recursive coordinates $M_\alpha(h)$.

The transition matrix B records the direct predecessor factors. Rank descent implies local nilpotence, so

$$M = (I - B)^{-1}v.$$

In earlier RRFS work the operator $(I - B)^{-1}$ was called the incidence resolvent. Here we emphasize a different interpretation and rename it the *Green operator*. Its entries sum all weighted recursive paths, and it converts a boundary source v into the full recursive population M .

The central observation of the present paper is dual. The transpose Green operator converts *local recursive energies* into the ordinary energies of root atoms. If ε_α is the energy attached to one occurrence of the recursive type α , then

$$\varepsilon^\top M(h) = \varepsilon^\top Gv(h) = (G^\top \varepsilon)^\top v(h).$$

Thus

recursive energy $\xrightarrow{G^\top}$ root energy.

Exponentiating turns the additive energy into a multiplicative norm. Summing the Boltzmann weights $e^{-sL(h)}$ over all objects then produces an analytic zeta function. In a factorial monoid the same function automatically factors as an Euler product over root atoms.

This gives the chain

$$B \longrightarrow G = (I - B)^{-1} \longrightarrow L_\varepsilon \longrightarrow \zeta_{\mathfrak{R},\varepsilon}(s) \longrightarrow \text{Euler product / prime laws.}$$

The Green operator is not itself the Riemann zeta function. Rather, it is the recursive propagator from which a family of analytic zeta functions is obtained after a gauge is chosen. The distinction is essential.

The main examples show that this framework recovers familiar zeta functions without changing them. For natural numbers one obtains $\zeta(s)$. For a number field K , the ideal-Pratt lift on nonzero ideals of $\mathcal{O}_K$ gives the Dedekind zeta function $\zeta_K(s)$ while replacing the numerical Pratt tree by a prime-ideal tree that records splitting and ramification. For monic polynomials over $\mathbb{F}_q$ one obtains $(1 - q^{1-s})^{-1}$. In the polynomial RRFS over $\mathbb{Q}$, Mahler measure yields a genuinely nontrivial recursive energy, but its global one-variable partition sum is obstructed by cyclotomic polynomials; fixed-degree and bivariate height zetas are therefore the natural objects. Terminal orbit families such as Jordan and Grassmann atoms supply a complementary limiting case in which $G = I$: the zeta function is nontrivial even though the recursive geometry is trivial.

2 RRFS background

2.1 Ranked recursive factorization systems

Definition 2.1 (Ranked recursive factorization system). *A ranked recursive factorization system is a quadruple*

$$\mathfrak{R} = (H, \mathcal{A}, D, \rho)$$

with the following properties.

- (i) H is a reduced factorial commutative cancellative monoid with atom set $\mathcal{A}$;
- (ii) $D : \mathcal{A} \rightarrow H$ is a predecessor map;
- (iii) $\rho : \mathcal{A} \rightarrow \mathbb{N}_0$ is a rank;
- (iv) if $\beta \mid D\alpha$, then $\rho(\beta) < \rho(\alpha)$.

An atom τ is terminal if $D\tau = 1$.

Every $h \in H$ has a unique finite factorization

$$h = \prod_{\alpha \in \mathcal{A}} \alpha^{v_\alpha(h)}.$$

Write

$$D\alpha = \prod_{\beta \in \mathcal{A}} \beta^{B_{\beta,\alpha}}, \quad B_{\beta,\alpha} = v_\beta(D\alpha).$$

The nonnegative matrix B is column-finite and locally nilpotent on finitely supported vectors.

For an atom α , unfold the predecessor rule recursively to obtain its finite atomic tree T_α . For a general object h , take $v_\alpha(h)$ copies of T_α . The total number of β -labelled vertices in this forest is denoted $M_\beta(h)$.

Theorem 2.2 (Recursive coordinate equation). *For every $h \in H$,*

$$M(h) = v(h) + BM(h),$$

and therefore

$$M(h) = (I - B)^{-1}v(h), \quad v(h) = (I - B)M(h).$$

Proof. Each occurrence of an atom is either a root occurrence, counted by v , or is forced as a direct child of some recursive occurrence, counted by BM . Rank descent makes the recursion finite. Rearranging gives the second pair of identities. $\square$

2.2 Telescoping reconstruction

Let $q(H)$ denote the group completion of H . Define

$$\Delta(\alpha) = \alpha(D\alpha)^{-1} \in q(H).$$

Theorem 2.3 (RRFS reconstruction). *For every $h \in H$,*

$$h = \prod_{\alpha \in \mathcal{A}} \Delta(\alpha)^{M_\alpha(h)} = \prod_{\alpha \in \mathcal{A}} \left(\frac{\alpha}{D\alpha} \right)^{M_\alpha(h)}.$$

Proof. The exponent of an atom β on the right is

$$M_\beta(h) - \sum_{\alpha} B_{\beta,\alpha} M_\alpha(h) = ((I - B)M(h))_\beta = v_\beta(h).$$

This is exactly the ordinary factorization exponent of β in h . $\square$

3 The Green operator

Definition 3.1 (Green operator). *The Green operator of an RRFS is*

$$G_{\mathfrak{R}} = (I - B)^{-1} = I + B + B^2 + \cdots,$$

where the series is interpreted locally: on each finitely supported vector only finitely many terms are nonzero.

The terminology reflects the source-to-population interpretation

$$M = G_{\mathfrak{R}}v.$$

A root multiplicity vector is a boundary source; the Green operator propagates it through every allowed recursive path.

Proposition 3.2 (Path expansion). *For atoms α, β ,*

$$(G_{\mathfrak{R}})_{\beta,\alpha} = \sum_{k \geq 0} \sum_{\alpha = \alpha_0 \rightarrow \alpha_1 \rightarrow \cdots \rightarrow \alpha_k = \beta} \prod_{j=1}^k B_{\alpha_j, \alpha_{j-1}}.$$

Thus $(G_{\mathfrak{R}})_{\beta,\alpha}$ is the total multiplicity of β in T_α .

3.1 Dual Green transport of energies

The key analytic use of $G_{\mathfrak{R}}$ is its transpose.

Let $\varepsilon = (\varepsilon_\alpha)$ be a real vector on atom types. Think of ε_α as the local energy of one recursive occurrence of α . Define

$$L_\varepsilon(h) = \sum_{\alpha} \varepsilon_\alpha M_\alpha(h).$$

Because recursive coordinates are additive under multiplication,

$$L_\varepsilon(hg) = L_\varepsilon(h) + L_\varepsilon(g).$$

Theorem 3.3 (Dual Green transform). *Let*

$$c = G_{\mathfrak{R}}^\top \varepsilon.$$

Then for every $h \in H$,

$$L_\varepsilon(h) = \varepsilon^\top G_{\mathfrak{R}} v(h) = c^\top v(h).$$

In particular, the energy of a root atom α is

$$c_\alpha = L_\varepsilon(\alpha).$$

Conversely,

$$\varepsilon = (I - B^\top)c,$$

so

$$\varepsilon_\alpha = c_\alpha - \sum_{\beta} B_{\beta,\alpha} c_\beta.$$

Proof. The first identity is

$$L_\varepsilon(h) = \varepsilon^\top M(h) = \varepsilon^\top G_{\mathfrak{R}} v(h) = (G_{\mathfrak{R}}^\top \varepsilon)^\top v(h).$$

Since $G_{\mathfrak{R}}^\top = (I - B^\top)^{-1}$, the inverse formula follows. $\square$

This is the analytic counterpart of boundary inversion. The ordinary root energy c_α equals the total local energy of the full recursive tree T_α .

3.2 Norms as Green energies

Definition 3.4 (Admissible Green gauge). *A local energy vector ε is called an admissible Green gauge if*

$$L_\varepsilon(h) = \varepsilon^\top G_{\mathfrak{R}} v(h) \geq 0$$

for all $h \in H$, $L_\varepsilon(h) = 0$ only at the unit (unless a controlled degeneracy is explicitly allowed), and bounded energy sets are finite whenever a proper counting theory is required.

Define

$$N_\varepsilon(h) = e^{L_\varepsilon(h)}.$$

Then N_ε is multiplicative.

Conversely, suppose a multiplicative norm $N : H \rightarrow \mathbb{R}_{>0}$ is given. Set

$$c_\alpha = \log N(\alpha).$$

Then the inverse dual Green transform gives the local recursive energy

$$\varepsilon_\alpha = \log N(\alpha) - \log N(D\alpha) = \log \frac{N(\alpha)}{N(D\alpha)}.$$

The reconstruction theorem immediately yields

$$\log N(h) = \sum_{\alpha} M_\alpha(h) \log \frac{N(\alpha)}{N(D\alpha)}.$$

Local energies may have either sign; positivity is a property of the total gauge, not of each recursive increment.

4 Zeta functions of recursive factorization systems

Definition 4.1 (RRFS zeta function). *Let ε be an admissible Green gauge. In every half-plane in which the series converges absolutely, define*

$$\zeta_{\mathfrak{R},\varepsilon}(s) = \sum_{h \in H} e^{-sL_\varepsilon(h)} = \sum_{h \in H} \exp\left(-s \varepsilon^\top G_{\mathfrak{R}} v(h)\right).$$

Equivalently, if $N = N_\varepsilon$,

$$\zeta_{\mathfrak{R},N}(s) = \sum_{h \in H} N(h)^{-s}.$$

The second notation emphasizes a pre-existing norm; the first emphasizes that the norm can be generated from local recursive energies by the Green operator.

Theorem 4.2 (Green-Euler duality). *Let $c = G_{\mathfrak{R}}^\top \varepsilon$, and suppose the relevant sums converge absolutely. Then*

$$\zeta_{\mathfrak{R},\varepsilon}(s) = \prod_{\alpha \in \mathcal{A}} (1 - e^{-sc_\alpha})^{-1}.$$

Equivalently, for a multiplicative norm N ,

$$\zeta_{\mathfrak{R},N}(s) = \prod_{\alpha \in \mathcal{A}} (1 - N(\alpha)^{-s})^{-1}.$$

Proof. Unique factorization identifies H with the finitely supported vectors $v \in \mathbb{N}_0^{(\mathcal{A})}$. Hence

$$\begin{aligned}\zeta_{\mathfrak{R},\varepsilon}(s) &= \sum_v e^{-sc^\top v} = \prod_{\alpha \in \mathcal{A}} \sum_{m \geq 0} e^{-smc_\alpha} \\ &= \prod_{\alpha \in \mathcal{A}} (1 - e^{-sc_\alpha})^{-1}.\end{aligned}$$

□

The theorem exhibits two coordinate descriptions of one function:

boundary coordinates v recursive coordinates $M = Gv$ $\longrightarrow$ Euler product, Green partition sum.
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The Euler product comes from factoriality. The recursive representation comes from the predecessor geometry.

4.1 Logarithmic derivative and the RRFS von Mangoldt weight

From the Euler product,

$$\log \zeta_{\mathfrak{R},N}(s) = \sum_{\alpha \in \mathcal{A}} \sum_{m \geq 1} \frac{N(\alpha)^{-ms}}{m}.$$

Differentiating termwise in a region of absolute convergence gives

$$-\frac{\zeta'_{\mathfrak{R},N}(s)}{\zeta_{\mathfrak{R},N}(s)} = \sum_{\alpha \in \mathcal{A}} \sum_{m \geq 1} (\log N(\alpha)) N(\alpha)^{-ms}.$$

Definition 4.3 (Boundary von Mangoldt weight). *Define*

$$\Lambda_{\mathfrak{R},N}(h) = \begin{cases} \log N(\alpha), & h = \alpha^m \text{ for some atom } \alpha \text{ and } m \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$-\frac{\zeta'_{\mathfrak{R},N}(s)}{\zeta_{\mathfrak{R},N}(s)} = \sum_{h \in H} \Lambda_{\mathfrak{R},N}(h) N(h)^{-s}.$

This weight is attached to the ordinary factorization boundary. It should not be confused with a Möbius transform taken in a Pratt-coordinate order. Both notions can coexist, and comparing them is an open analytic problem.

4.2 Direct sums

If two RRFS have disjoint atom sets and independent gauges, their direct-sum monoid has the product norm. Consequently

$$\zeta_{\mathfrak{R}_1 \oplus \mathfrak{R}_2}(s) = \zeta_{\mathfrak{R}_1}(s) \zeta_{\mathfrak{R}_2}(s).$$

This mirrors the block-diagonal identity

$$G_{\mathfrak{R}_1 \oplus \mathfrak{R}_2} = G_{\mathfrak{R}_1} \oplus G_{\mathfrak{R}_2}.$$

5 Natural numbers: the Riemann zeta function

Let

$$H = \mathbb{N}, \quad \mathcal{A} = \{2, 3, 5, \dots\},$$

with the usual prime factorization. Define

$$D(2) = 1, \quad D(p) = p - 1 \quad (p > 2).$$

The rank may be taken to increase with the prime, since every prime divisor of $p - 1$ is smaller than p . The transition matrix is

$$B_{q,p} = v_q(p - 1).$$

The recursive coordinates are the numerical Pratt coordinates $m_q(n)$:

$$M(n) = Gv(n), \quad G = (I - B)^{-1}.$$

Take the ordinary norm $N(n) = n$. Root energies are

$$c_p = \log p.$$

The inverse dual Green transform gives

$$\varepsilon_p = c_p - \sum_q v_q(p - 1)c_q = \log p - \log(p - 1) = \boxed{\log \frac{p}{p - 1}}.$$

Therefore

$$\boxed{\log n = \sum_p m_p(n) \log \frac{p}{p - 1}}.$$

This is exactly the logarithmic Pratt reconstruction formula.

The RRFS zeta function is consequently

$$\begin{aligned} \zeta_{\mathfrak{R}}(s) &= \sum_{n \geq 1} \exp \left(-s \sum_p m_p(n) \log \frac{p}{p - 1} \right) \\ &= \sum_{n \geq 1} n^{-s}. \end{aligned}$$

Hence

$$\boxed{\zeta_{\mathfrak{R}}(s) = \zeta(s)} \quad (\Re s > 1).$$

The Green-Euler theorem simultaneously gives

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}.$$

This example is the basic model for the paper. The classical zeta function is unchanged; what is new is a second coordinate representation of its energy:

$$\boxed{\log n = (\log p)_p^{\mathsf{T}} v(n) = \left(\log \frac{p}{p - 1} \right)_p^{\mathsf{T}} Gv(n)}.$$

The first side is boundary geometry, the second is Pratt geometry.

The logarithmic derivative recovers the classical von Mangoldt function:

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}.$$

Thus the ordinary PNT and RH remain the analytic statements attached to the same RRFS zeta.

6 Dedekind domains and ideal-Pratt geometry

The numerical Pratt predecessor can be lifted from rational primes to prime ideals in a number field. This produces a nonterminal RRFS whose analytic zeta function is the classical Dedekind zeta function, but whose Green operator refines the numerical Pratt geometry by ideal splitting and ramification. The construction uses ideal factorization rather than factorization of elements, so it applies even when $\mathcal{O}_K$ is not a unique factorization domain.

6.1 The ideal monoid

Let $K/\mathbb{Q}$ be a number field of degree

$$n = [K : \mathbb{Q}],$$

and let $\mathcal{O}_K$ be its ring of integers. Write

$$H_K = \{\mathfrak{a} \subseteq \mathcal{O}_K : \mathfrak{a} \neq 0\}$$

for the multiplicative monoid of nonzero integral ideals. Every such ideal has a unique factorization

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\mathfrak{a})}$$

into nonzero prime ideals. Thus H_K is a factorial commutative monoid with atom set

$$\mathcal{A}_K = \{\mathfrak{p} \subset \mathcal{O}_K : \mathfrak{p} \neq 0 \text{ prime}\}.$$

The multiplicative norm is

$$N\mathfrak{a} = |\mathcal{O}_K/\mathfrak{a}|, \quad N(\mathfrak{a}\mathfrak{b}) = N\mathfrak{a} N\mathfrak{b}.$$

For a nonzero prime ideal $\mathfrak{p}$, its contraction to $\mathbb{Z}$ is

$$\mathfrak{p} \cap \mathbb{Z} = (p)$$

for a unique rational prime p . We call p the rational prime below $\mathfrak{p}$.

6.2 The ideal-Pratt predecessor

Definition 6.1 (Ideal-Pratt RRFS). *For $\mathfrak{p} \mid p$ define*

$$D_K(\mathfrak{p}) = (p-1)\mathcal{O}_K.$$

For prime ideals over 2 this is the unit ideal. Give $\mathfrak{p} \mid p$ the rank

$$\rho_K(\mathfrak{p}) = p,$$

or, equivalently, any rank that strictly increases with the underlying rational prime.

Because $\mathcal{O}_K$ is Dedekind, every predecessor has a finite prime-ideal factorization

$$(p-1)\mathcal{O}_K = \prod_{\mathfrak{q}} \mathfrak{q}^{B_{\mathfrak{q},p}^K}.$$

This defines the transition matrix B_K .

Proposition 6.2 (Rank descent). *The quadruple*

$$\mathfrak{R}_K^{\text{Pratt}} = (H_K, \mathcal{A}_K, D_K, \rho_K)$$

is a ranked recursive factorization system.

Proof. Only rank descent needs verification. If

$$\mathfrak{q} \mid D_K(\mathfrak{p}) = (p-1)\mathcal{O}_K$$

and $\mathfrak{q} \mid q$ lies above the rational prime q , then $q \mid p-1$. Hence $q < p$, so

$$\rho_K(\mathfrak{q}) < \rho_K(\mathfrak{p}).$$

Finite ideal factorization gives finite branching. Therefore repeated predecessors terminate at prime ideals above 2. $\square$

More explicitly, if $\mathfrak{p} \mid p$ and $\mathfrak{q} \mid q$, then

$$B_{\mathfrak{q},\mathfrak{p}}^K = e(\mathfrak{q}/q) v_q(p-1),$$

where $e(\mathfrak{q}/q)$ is the ramification index of $\mathfrak{q}$ over q . Thus the numerical transition multiplicity $v_q(p-1)$ is lifted to every prime ideal above q and multiplied by its ramification index.

The Green operator is

$$G_K = (I - B_K)^{-1}.$$

For a nonzero ideal $\mathfrak{a}$ its ideal-Pratt coordinates are

$$M_K(\mathfrak{a}) = G_K v(\mathfrak{a}).$$

The entry $(G_K)_{\mathfrak{q},\mathfrak{p}}$ counts, with factorization multiplicities, the total number of occurrences of $\mathfrak{q}$ in the fully unfolded ideal-Pratt tree rooted at $\mathfrak{p}$.

6.3 Green reconstruction and the Dedekind zeta function

The abstract RRFS reconstruction theorem gives an identity in the group of fractional ideals:

$$\mathfrak{a} = \prod_{\mathfrak{p}} \left(\frac{\mathfrak{p}}{(p-1)\mathcal{O}_K} \right)^{M_{\mathfrak{p}}(\mathfrak{a})}.$$

Here and below p denotes the rational prime below $\mathfrak{p}$.

If $f_{\mathfrak{p}} = f(\mathfrak{p}/p)$ is the residue degree, then

$$N\mathfrak{p} = p^{f_{\mathfrak{p}}}$$

and

$$N((p-1)\mathcal{O}_K) = (p-1)^n.$$

Hence the local recursive energy attached to $\mathfrak{p}$ is

$$\varepsilon_{\mathfrak{p}} = \log \frac{N\mathfrak{p}}{N(D_K\mathfrak{p})} = f_{\mathfrak{p}} \log p - n \log(p-1).$$

The sign of $\varepsilon_{\mathfrak{p}}$ need not be positive. Nevertheless the global identity is exact:

$$\log N\mathfrak{a} = \sum_{\mathfrak{p}} M_{\mathfrak{p}}(\mathfrak{a}) (f_{\mathfrak{p}} \log p - n \log(p-1)).$$

Equivalently, if $c_{\mathfrak{p}} = \log N\mathfrak{p}$, then

$$c = G_K^{\top} \varepsilon.$$

The residue degrees therefore enter through the root-energy vector c , while ramification enters the Green operator through B_K .

Theorem 6.3 (Dedekind zeta as an RRFS zeta). *For $\Re s > 1$,*

$$\zeta_{\mathfrak{R}_K^{\text{Pratt}, \varepsilon}}(s) = \sum_{\mathfrak{a} \neq 0} (N\mathfrak{a})^{-s} = \zeta_K(s).$$

Equivalently,

$$\zeta_K(s) = \sum_{\mathfrak{a} \neq 0} \exp(-s \varepsilon^{\top} G_K v(\mathfrak{a})) = \prod_{\mathfrak{p}} (1 - (N\mathfrak{p})^{-s})^{-1}.$$

Proof. The Green reconstruction gives

$$\varepsilon^{\top} G_K v(\mathfrak{a}) = \log N\mathfrak{a}.$$

Substitution in the RRFS partition sum gives the defining Dirichlet series of the Dedekind zeta function. Unique factorization of ideals gives the Euler product. $\square$

Thus the Dedekind zeta function has two simultaneous coordinate descriptions. The Euler product sees ordinary prime-ideal factorization. The Green representation sees the same norm after propagating through the lifted Pratt recursion.

6.4 A general Dedekind lift

The preceding construction is a special case of a general extension principle.

Proposition 6.4 (Dedekind lifting of an RRFS). *Let $A \subseteq B$ be a finite extension of Dedekind domains. Suppose the monoid of nonzero ideals of A carries a ranked RRFS with predecessor D_A on nonzero prime ideals. For a prime ideal $\mathfrak{P}$ of B with contraction $\mathfrak{p} = \mathfrak{P} \cap A$, define*

$$D_B(\mathfrak{P}) = D_A(\mathfrak{p})B.$$

Assign $\mathfrak{P}$ the rank of $\mathfrak{p}$. Then this defines a ranked RRFS on the nonzero ideals of B .

Proof. Factor $D_A(\mathfrak{p})$ in A . Every prime ideal $\mathfrak{Q}$ dividing its extension to B lies above a prime $\mathfrak{q}$ dividing $D_A(\mathfrak{p})$. Rank descent in A therefore implies rank descent in B . Unique ideal factorization supplies finite branching. $\square$

In transition-matrix form, if $\mathfrak{P} \mid \mathfrak{p}$ and $\mathfrak{Q} \mid \mathfrak{q}$, then

$$B_{\mathfrak{Q}, \mathfrak{P}}^B = e(\mathfrak{Q}/\mathfrak{q}) B_{\mathfrak{q}, \mathfrak{p}}^A.$$

The ideal-Pratt construction for $\mathcal{O}_K/\mathbb{Z}$ is obtained by taking

$$D_{\mathbb{Z}}((p)) = (p - 1).$$

6.5 Galois symmetry

Assume now that $K/\mathbb{Q}$ is Galois. The Galois group acts on prime ideals and on the ideal factorization space. Since every automorphism fixes rational integers,

$$D_K(\sigma \mathfrak{p}) = (p - 1) \mathcal{O}_K = \sigma D_K(\mathfrak{p}).$$

Consequently

$$B_{\sigma \mathfrak{q}, \sigma \mathfrak{p}}^K = B_{\mathfrak{q}, \mathfrak{p}}^K$$

and therefore the permutation action of $\text{Gal}(K/\mathbb{Q})$ commutes with B_K and with G_K .

Proposition 6.5 (Galois equivariance). *For a finite Galois extension $K/\mathbb{Q}$, the ideal-Pratt Green operator G_K is $\text{Gal}(K/\mathbb{Q})$ -equivariant.*

This makes it natural to ask whether representation-theoretic decompositions of Green data can be coupled to characters or Frobenius weights. We do not identify such constructions with Artin L -functions here; this is left as an analytic problem.

6.6 Worked example: Gaussian integers

Let $K = \mathbb{Q}(i)$ and $\mathcal{O}_K = \mathbb{Z}[i]$. Write

$$\mathfrak{p}_2 = (1 + i), \quad \mathfrak{p}_3 = (3).$$

The prime 2 ramifies,

$$(2) = \mathfrak{p}_2^2,$$

while 3 is inert, so $\mathfrak{p}_3 = (3)$ is prime with norm 9. The rational prime 13 splits; let $\mathfrak{p}_{13}$ be either prime ideal above 13. Then

$$D_K(\mathfrak{p}_{13}) = (12) = (2)^2(3) = \mathfrak{p}_2^4 \mathfrak{p}_3,$$

and

$$D_K(\mathfrak{p}_3) = (2) = \mathfrak{p}_2^2, \quad D_K(\mathfrak{p}_2) = \mathcal{O}_K.$$

Thus the full recursive population of the tree rooted at $\mathfrak{p}_{13}$ is

$$M_{\mathfrak{p}_{13}} = 1, \quad M_{\mathfrak{p}_3} = 1, \quad M_{\mathfrak{p}_2} = 6.$$

The local energies are

$$\varepsilon_{\mathfrak{p}_{13}} = \log \frac{13}{12^2}, \quad \varepsilon_{\mathfrak{p}_3} = \log \frac{9}{2^2}, \quad \varepsilon_{\mathfrak{p}_2} = \log 2.$$

They telescope exactly:

$$\log 13 = \log \frac{13}{144} + \log \frac{9}{4} + 6 \log 2.$$

This small example displays all three arithmetic ingredients at once: splitting at 13, inertia at 3, and ramification at 2.

6.7 Same zeta, different Green geometry

There is also a terminal RRFS on the same ideal monoid, obtained by declaring

$$D_{\text{term}}(\mathfrak{p}) = \mathcal{O}_K.$$

Its Green operator is the identity, yet with the same ideal norm its analytic zeta function is again $\zeta_K(s)$. Therefore the Dedekind example gives an immediate and concrete instance of an important general phenomenon:

the analytic zeta function does not determine the Green operator.

The terminal and ideal-Pratt RRFS have the same atom monoid, the same norm, and the same Dedekind zeta function, but different recursive geometries. In this precise sense the zeta function can be viewed as an analytic shadow of finer Green data.

7 Polynomial RRFS

7.1 The polynomial predecessor

Let H be the multiplicative monoid of monic polynomials in $\mathbb{Q}[x]$, with monic irreducibles as atoms. For an irreducible ϕ , define

$$D\phi = \phi - \phi(0), \quad Dx = 1.$$

Since $D\phi$ has zero constant term, x divides it; after removing x , every remaining irreducible factor has smaller degree than ϕ . This yields a finite recursive forest and the usual polynomial Pratt coordinates $M_\psi(f)$.

The reconstruction theorem reads

$$f = \prod_{\phi} \left(\frac{\phi}{D\phi} \right)^{M_\phi(f)}.$$

A distinguished terminal-coordinate theorem gives

$$M_x(f) = \deg f.$$

7.2 Finite fields and the affine-line zeta function

The same predecessor construction works over $\mathbb{F}_q[x]$ for monic irreducible polynomials, with x terminal. Take the degree norm

$$N_q(f) = q^{\deg f}.$$

For $\phi \neq x$,

$$\deg D\phi = \deg \phi,$$

so

$$\varepsilon_\phi = \log \frac{N_q(\phi)}{N_q(D\phi)} = 0.$$

For the terminal atom x ,

$$\varepsilon_x = \log q.$$

Therefore the complete recursive energy collapses to one coordinate:

$$\log N_q(f) = (\log q)M_x(f) = (\log q) \deg f.$$

The RRFS zeta function is

$$\begin{aligned} \zeta_{\mathfrak{R},q}(s) &= \sum_{f \text{ monic}} q^{-s \deg f} \\ &= \sum_{n \geq 0} q^n q^{-sn} = \boxed{\frac{1}{1 - q^{1-s}}}. \end{aligned}$$

Equivalently,

$$\boxed{\frac{1}{1 - q^{1-s}} = \prod_{P \text{ monic irreducible}} \left(1 - q^{-s \deg P}\right)^{-1}}.$$

This is the usual zeta function of the affine line over $\mathbb{F}_q$, written in the variable s .

The Green interpretation adds a precise statement: the degree gauge is blind to every nonterminal Pratt coordinate. All recursive energy flows to the terminal type x .

The classical exact identity

$$q^n = \sum_{d|n} d I_q(d)$$

and its Möbius inversion

$$I_q(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}$$

are therefore analytic consequences associated with the same zeta function and its Euler product.

7.3 Mahler measure over $\mathbb{Q}$

For a monic polynomial

$$f(x) = \prod_{j=1}^d (x - \alpha_j) \in \mathbb{Z}[x],$$

its Mahler measure is

$$\mathcal{M}(f) = \prod_{j=1}^d \max\{1, |\alpha_j|\}.$$

It is multiplicative. Hence root energies are

$$c_\phi = \log \mathcal{M}(\phi),$$

and the inverse dual Green transform gives the local Mahler increments

$$\varepsilon_\phi^{\mathcal{M}} = \log \frac{\mathcal{M}(\phi)}{\mathcal{M}(D\phi)}.$$

Therefore

$$\log \mathcal{M}(f) = \sum_{\phi} M_\phi(f) \log \frac{\mathcal{M}(\phi)}{\mathcal{M}(D\phi)}.$$

Unlike the numerical Pratt energies, the local Mahler increments need not be positive. The total energy remains $\log \mathcal{M}(f) \geq 0$.

There is also a global degeneracy: infinitely many cyclotomic irreducibles have Mahler measure 1. Thus

$$\sum_f \mathcal{M}(f)^{-s}$$

is not a proper one-variable zeta function over all degrees.

Two natural repairs preserve the Mahler gauge.

First, fix the degree:

$$Z_{\mathcal{M},d}(s) = \sum_{\substack{f \in \mathbb{Z}[x] \text{ monic} \\ \deg f = d}} \mathcal{M}(f)^{-s}.$$

Chern and Vaaler's lattice-point asymptotics for monic Mahler balls give growth of order T^d for fixed d . Consequently the natural abscissa for this height Dirichlet series is $\Re s = d$ (more precisely, absolute convergence holds for $\Re s > d$, while the counting law forces divergence at and below the critical growth exponent).

Second, retain all degrees but add the degree gauge:

$$Z_{\mathcal{M}}(s, t) = \sum_{f \text{ monic}} \mathcal{M}(f)^{-s} e^{-t \deg f},$$

in its domain of absolute convergence. Unique factorization gives

$$Z_{\mathcal{M}}(s, t) = \prod_{\phi \text{ monic irreducible}} \left(1 - \mathcal{M}(\phi)^{-s} e^{-t \deg \phi}\right)^{-1}.$$

The Green-coordinate expression is especially transparent because

$$\deg f = M_x(f) :$$

$$Z_{\mathcal{M}}(s, t) = \sum_f \exp\left(-s(\varepsilon^{\mathcal{M}})^{\top} Gv(f) - t e_x^{\top} Gv(f)\right).$$

Thus Mahler geometry and terminal-degree geometry become two independent linear functionals on the same Pratt coordinate space.

8 A finite Green example and engineered RRFS

The Green formalism is not restricted to arithmetic.

Example 8.1 (A three-level chain). *Let the atom set be $\{a_0, a_1, a_2\}$ and define*

$$Da_0 = 1, \quad Da_1 = a_0^r, \quad Da_2 = a_1^s,$$

with $r, s \geq 1$. In the ordered basis (a_0, a_1, a_2) ,

$$B = \begin{pmatrix} 0 & r & 0 \\ 0 & 0 & s \\ 0 & 0 & 0 \end{pmatrix}, \quad G = (I - B)^{-1} = \begin{pmatrix} 1 & r & rs \\ 0 & 1 & s \\ 0 & 0 & 1 \end{pmatrix}.$$

If the local recursive energies are $\varepsilon = (\varepsilon_0, \varepsilon_1, \varepsilon_2)^\top$, then

$$c = G^\top \varepsilon = \begin{pmatrix} \varepsilon_0 \\ r\varepsilon_0 + \varepsilon_1 \\ rs\varepsilon_0 + s\varepsilon_1 + \varepsilon_2 \end{pmatrix}.$$

Hence

$$\zeta_{\mathfrak{A}}(s) = \prod_{j=0}^2 (1 - e^{-sc_j})^{-1}.$$

The energy of a higher atom is literally the total local energy of its recursive descendant tree.

A finite dependency RRFS, bill of materials, attack tree, or software package system behaves in the same way. Assign an energy to each recursive occurrence (cost, risk, code size, or another additive quantity). The Green operator propagates the local energies to root-product energies, and a finite atom set gives a finite Euler product. Such zeta functions are analytic generating functions rather than arithmetic zeta functions; no prime-number interpretation should be asserted without an additional asymptotic family.

9 Terminal RRFS from orbit atom families

Jordan and Grassmann systems were introduced naturally as Orbit Systems rather than nontrivial RRFS. Nevertheless, their prime-field orbit atoms can be placed in an RRFS in a conservative way: declare every orbit atom terminal. This produces a legitimate RRFS with

$$B = 0, \quad G = I.$$

The construction adds no recursive geometry. Its value is analytic: it separates the zeta effect of atom multiplicity and norm from the genuinely recursive effect of a nonzero Green operator.

9.1 A general geometric terminalization

Proposition 9.1 (Dimension-balanced terminal zeta). *Suppose that for every rational prime p we have a finite atom set A_p with*

$$|A_p| = p^D + O(p^{D-1})$$

for some integer $D \geq 0$. Form the free commutative monoid on the disjoint union $\bigsqcup_p A_p$, declare every atom terminal, and assign each $a \in A_p$ the norm

$$N(a) = p^{D+1}.$$

Then

$$Z_D(s) = \prod_p \left(1 - p^{-(D+1)s}\right)^{-|A_p|}.$$

Moreover, in a neighborhood of $s = 1$,

$$\boxed{Z_D(s) = \zeta((D+1)s - D) H_D(s),}$$

where H_D is holomorphic and nonzero near $s = 1$.

Proof. Take logarithms. The $m = 1$ term is

$$\sum_p |A_p| p^{-(D+1)s} = \sum_p p^{-((D+1)s-D)} + O\left(\sum_p p^{-((D+1)s-D+1)}\right).$$

The error converges near $s = 1$. Terms with $m \geq 2$ are also absolutely convergent in a neighborhood of 1. The same decomposition for $\log \zeta((D+1)s - D)$ shows that the difference of the two logarithms is holomorphic there. Exponentiation yields the factorization. $\square$

The atom counting function, measured by the norm cutoff, satisfies

$$\Pi_D(X) = \sum_{p^{D+1} \leq X} |A_p| \sim \frac{X}{\log X}$$

by the classical PNT and partial summation. The balancing exponent $D + 1$ converts p^D geometric multiplicity into an ordinary prime-number scale.

9.2 Jordan orbit atoms

For fixed $k \geq 1$, the Jordan orbit atoms over p are the scalar orbits of nonzero vectors in $\mathbb{F}_p^k$, equivalently

$$\mathbb{P}^{k-1}(\mathbb{F}_p).$$

Their number is

$$a_k(p) = \frac{p^k - 1}{p - 1} = 1 + p + \cdots + p^{k-1} = p^{k-1} + O(p^{k-2}).$$

Here $D = k - 1$. Assign norm

$$N(a) = p^k.$$

The terminal Jordan zeta is

$$\boxed{Z_{J,k}(s) = \prod_p \left(1 - p^{-ks}\right)^{-(1+p+\cdots+p^{k-1})}.$$

The preceding proposition gives

$$\boxed{Z_{J,k}(s) = \zeta(ks - k + 1)H_{J,k}(s)}$$

near $s = 1$, with $H_{J,k}$ holomorphic and nonzero there. For $k = 1$ this is exactly the Riemann zeta function.

The Green operator is I , so this shifted Riemann-zeta singularity comes from projective atom multiplicity and the chosen dimension-balanced norm, not from recursive propagation.

9.3 Grassmann orbit atoms

Fix $1 \leq r < k$ and let

$$D = r(k - r) = \dim \operatorname{Gr}(r, k).$$

At a prime p , the Grassmann atom set is

$$\operatorname{Gr}(r, k)(\mathbb{F}_p),$$

whose cardinality is the Gaussian binomial coefficient

$$\left[\begin{matrix} k \\ r \end{matrix} \right]_p = p^D + O(p^{D-1}).$$

Declare these atoms terminal and assign norm

$$N(a) = p^{D+1}.$$

Then

$$\boxed{Z_{k,r}^{\operatorname{Gr}}(s) = \prod_p \left(1 - p^{-(D+1)s} \right)^{-\left[\begin{matrix} k \\ r \end{matrix} \right]_p}$$

and

$$\boxed{Z_{k,r}^{\operatorname{Gr}}(s) = \zeta((D+1)s - D)H_{k,r}(s)}$$

near $s = 1$.

Again this is a terminal RRFS, not a claim that the Grassmann Orbit System possesses a canonical nontrivial predecessor map. Finding such a predecessor, if one exists, would produce a genuinely new Green operator rather than the identity.

10 Prime number laws, Orbit Systems, and zeros

The zeta formalism and the Orbit-System formalism emphasize different operations.

The RRFS zeta begins with the factorial atom boundary and packages it analytically through

$$\zeta_{\mathfrak{A},N}(s) = \prod_{\alpha} (1 - N(\alpha)^{-s})^{-1}.$$

Its logarithmic derivative isolates atom powers through $\Lambda_{\mathfrak{A},N}$.

An Orbit System begins with a locally finite type poset, primitive strata, symmetry quotients, and a gauge. Möbius inversion in that poset isolates primitive mass. A Pratt-order Möbius transform can therefore differ from the boundary von Mangoldt function even when both are built from the same RRFS.

The natural bridge is analytic. Once an OS counting function can be represented by a logarithmic derivative, Mellin transform, or coefficient extraction from an RRFS zeta, poles control main terms and zeros control oscillation. This is the familiar mechanism for the classical PNT and RH, now available as a template for other RRFS/OS pairs.

System	Green data	Zeta function	Leading analytic feature
Natural numbers	$B_{q,p} = v_q(p-1)$	$\zeta(s)$	pole at $s = 1$
Dedekind ideals	$D_K(\mathfrak{p}) = (p-1)\mathcal{O}_K$	$\zeta_K(s)$	prime-ideal Euler factors
Finite-field polynomials	polynomial B ; degree energy terminal	$(1 - q^{1-s})^{-1}$	pole at $s = 1$
Mahler polynomial RRFS	nontrivial G , signed Mahler increments	$Z_{\mathcal{M},d}(s)$ or $Z_{\mathcal{M}}(s, t)$	height-growth boundary
Jordan terminal RRFS	$G = I$	$Z_{J,k}(s)$	$\zeta(k s - k + 1)$ factor
Grassmann terminal RRFS	$G = I$	$Z_{k,r}^{\text{Gr}}(s)$	$\zeta((D+1)s - D)$ factor

The table also clarifies an important point: a nontrivial zeta function does not require a nontrivial Green operator. The zeta function combines three independent structures:

factorial atom set + recursive Green geometry + analytic gauge.

Different examples may place the complexity in different factors.

11 Open problems

Problem 11.1 (Analytic continuation from Green data). *Find structural conditions on (B, ε) that imply meromorphic continuation of*

$$\zeta_{\mathfrak{R}, \varepsilon}(s) = \sum_h e^{-s \varepsilon^{\text{T}} G v(h)}.$$

The Euler product reduces the question to the transported root energies $c = G^{\text{T}} \varepsilon$, but it is desirable to recognize continuation directly from recursive geometry.

Problem 11.2 (Green explicit formula). *Develop an explicit-formula formalism linking zeros and poles of $\zeta_{\mathfrak{R}, \varepsilon}$ to weighted primitive counting functions and, when present, to Möbius-primitive masses in a Pratt-order Orbit System.*

Problem 11.3 (Mahler-Pratt zeta). *Study the analytic continuation, singularities, and coefficient asymptotics of*

$$Z_{\mathcal{M}}(s, t) = \sum_f \mathcal{M}(f)^{-s} e^{-t \deg f}$$

and its fixed-degree sections. Determine which features of the singular set are visible in the polynomial Green operator rather than only in the ordinary irreducible Euler product.

Problem 11.4 (Nonterminal geometric predecessors). *The Jordan and Grassmann constructions in this paper are terminal. Find natural rank-decreasing predecessors on their orbit atoms, or prove that no canonical choice exists under reasonable functoriality requirements. A nonterminal construction would produce new Green operators and genuinely recursive geometric zeta functions.*

Problem 11.5 (Arithmetic equivalence versus Green equivalence). *The terminal and ideal-Pratt RRFS on $\mathcal{O}_K$ have the same Dedekind zeta function but different Green operators. More generally, determine which arithmetic information is retained by G_K but lost by ζ_K . In particular, for arithmetically equivalent number fields with the same Dedekind zeta function, compare their ideal-Pratt Green operators and ask whether an appropriate notion of Green equivalence is strictly finer than zeta equivalence.*

Problem 11.6 (Galois Green decomposition and L -functions). *For a finite Galois extension $K/\mathbb{Q}$, decompose the $\text{Gal}(K/\mathbb{Q})$ -equivariant Green operator into isotypic components. Determine whether natural character-weighted partition sums or Green traces recover known Artin L -functions, or define genuinely different recursive refinements. No such identification is asserted here.*

Problem 11.7 (Zeta functoriality). *Determine which RRFS morphisms induce identities, divisibility relations, or transfer maps between zeta functions. Direct sums already give products; extensions and quotients should have more subtle analytic behavior.*

Problem 11.8 (Square-root laws). *For RRFS zeta functions admitting a prime-number law, formulate conditions under which the remainder is governed by a square-root scale. In classical arithmetic this is controlled by zeros on a critical line; in general RRFS it is not yet clear which part of the structure plays the analogous role.*

12 Conclusion

The Green operator

$$G_{\mathfrak{R}} = (I - B)^{-1}$$

is the natural propagator of a recursive factorization system. It converts root multiplicities into total recursive populations. Its transpose performs the dual operation on energies: local recursive increments ε are transported to root energies

$$c = G_{\mathfrak{R}}^{\top} \varepsilon.$$

This produces a multiplicative norm

$$N_{\varepsilon}(h) = \exp(\varepsilon^{\top} G_{\mathfrak{R}} v(h))$$

and hence the RRFS zeta function

$$\boxed{\zeta_{\mathfrak{R},\varepsilon}(s) = \sum_h \exp(-s \varepsilon^{\top} G_{\mathfrak{R}} v(h)) = \prod_{\alpha} (1 - e^{-s(G_{\mathfrak{R}}^{\top} \varepsilon)_{\alpha}})^{-1}.}$$

This identity is the central bridge of the paper. The left side is written in recursive Pratt geometry; the right side is written in ordinary factorization geometry.

For the numerical Pratt RRFS the bridge recovers the Riemann zeta function from the positive local energies $\log(p/(p-1))$. For the nonzero ideals of $\mathcal{O}_K$, the lifted predecessor $D_K(\mathfrak{p}) = (p-1)\mathcal{O}_K$ produces a nontrivial ideal-Pratt Green operator while the analytic zeta remains the classical Dedekind zeta function $\zeta_K(s)$. This example also shows concretely that a zeta function need not determine its underlying Green geometry: the terminal and ideal-Pratt RRFS on the same ideal monoid have the same ζ_K but different Green operators. For finite-field polynomials the degree gauge collapses to the terminal coordinate and gives the affine-line zeta function. For rational polynomials, Mahler measure produces a nontrivial signed recursive energy and a natural family of height zeta functions. Jordan and Grassmann orbit atoms fit as terminal RRFS whose Green operator is the identity and whose zeta singularities arise from geometric atom multiplicities.

The resulting perspective is deliberately modest about classical analysis but ambitious about structure. The zeta function is not inserted independently of the recursive system. It is generated by coupling a Green geometry with a gauge. This suggests an analytic program for RRFS parallel to the classical sequence

$$\text{Euler product} \rightarrow \text{logarithmic derivative} \rightarrow \text{PNT} \rightarrow \text{explicit formula} \rightarrow \text{RH},$$

with the Green operator providing the additional recursive coordinate layer absent from ordinary analytic number theory.

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